

Probability and statistics - 1st semester - Final

March 31, 2026

Please detail your reasoning as much as possible : write the formulas, write the computations (even without a calculator) as much as you can; this will help you get points even without the full answer.

Use of phones, documents, or any form of cheating is strictly prohibited. Any student caught cheating will receive a grade of 0.

In total, there are 7 exercises for a total of 22 points

Cheatsheet and mathematical hints can be found at the end of the document.

Exercise 1 : lecture questions (2.5 pts)

0.5 pts per question

1. What is the difference between discrete and continuous random variables ?
2. What is the range of possible values for the correlation ?
3. What is the range of possible values for the covariance ?
4. What does the law of large numbers state ?
5. When are two random variables independent?

Exercise 2 : Uniform distribution (1.5 pts)

Let X be a continuous random variable with probability density function

$$f(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b, \\ 0, & \text{otherwise,} \end{cases}$$

where $a < b$ are real numbers.

1. Show that f is a probability density function.
2. Compute the expectation $E[X]$.
3. Compute the variance $\text{Var}(X)$.

Exercise 3 : continuous random variables (5 pts)

Let X be a continuous random variable supported on $[1, 2]$ with density

$$f_X(x) = \begin{cases} 2k x^2, & 1 \leq x \leq 2, \\ 0, & \text{otherwise,} \end{cases}$$

where $k > 0$ is a constant.

1. Determine the value of k so that f_X is a valid probability density function. (1 pt)
2. Find $F_X(x)$, the cdf of X . (0.75 pt)
3. Compute $P(X < 3/2)$ (0.25 pt)
4. Compute $E[X]$ (1 pt)
5. Compute $E[X^2]$ and $\text{Var}(X)$ (1 pt)

Exercise 4 : Joint Distribution, Covariance, and Correlation (4 pts)

Let U and V be discrete random variables with the following marginal distributions:

$$P(U = 2) = P(U = -2) = \frac{1}{2}, \quad P(V = 3) = P(V = -3) = \frac{1}{2}.$$

Define

$$d = P(U = 2 \text{ and } V = 3).$$

1. Express the **joint distribution** of U and V in terms of d (1 pt).
2. Compute the **covariance** $\text{Cov}(U, V)$ as a function of d (1 pt).
3. Compute the **correlation** $\text{Cor}(U, V)$ as a function of d (1 pt).

4. For which value(s) of d are U and V **independent** (0.5 pt)?
5. For which value(s) of d are U and V **perfectly correlated** (0.5 pt)?

Exercise 5 : inferential statistics (5 pts)

Carbon monoxide (CO) emissions (in gm/mi) for a certain kind of car vary with mean

$$\mu = 3.0$$

and standard deviation

$$\sigma = 0.45.$$

A company owns 64 cars of this type, acquired from various (i.e., random) sources. Assume that the CO emissions of different cars are independent and identically distributed.

1. What is the probability that a randomly selected car from the fleet has CO emissions greater than 3.2 gm/mi? (0.75 pt)
2. What is the probability that the average CO emissions of the 64 cars is greater than 3.2 gm/mi? (0.75 pt)
3. There is only a 1% chance that the fleet's mean CO emissions exceeds what value? (0.75 pt)
4. If the distribution of individual car emissions is not normal, explain why the approximation used in Question 2 remains valid. (0.5 pt)

Exercise 6 : MoM and MLE (5 pts)

Let X be a continuous random variable with probability density function

$$f(x; \theta) = \begin{cases} \frac{1}{\theta} e^{-(x-1)/\theta}, & x \geq 1, \\ 0, & \text{otherwise,} \end{cases}$$

where $\theta > 0$ is an unknown parameter.

A random sample of size $n = 10$ is observed:

$$x_1 = 1.42, x_2 = 1.35, x_3 = 1.58, x_4 = 1.26, x_5 = 1.47, \\ x_6 = 1.31, x_7 = 1.62, x_8 = 1.39, x_9 = 1.51, x_{10} = 1.44.$$

Method of Moment (2 pts)

1. Compute the theoretical $E[X]$ as a function of θ using the integral
Hint: Use the substitution $u = x - 1$ to reduce the integral to a sum of standard exponential integrals.
2. Derive the Method of Moments estimator $\hat{\theta}_{\text{MoM}}$.
3. Compute the numerical value of $\hat{\theta}_{\text{MoM}}$ using the observed data.
4. Is your estimator biased?

Maximum Likelihood Estimator (2 pts)

1. Write the likelihood function $L(\theta)$ based on the sample.
2. Find the Maximum Likelihood Estimator $\hat{\theta}_{\text{MLE}}$.
Hint: Use the log-likelihood.
3. Compute its numerical value using the observed sample.
4. Is your estimator biased?

Conclusions (1 pt)

1. State which estimator is the best and justify.

Exercise 7 : Discrete random variables (2 pts)

A card is drawn uniformly at random from a deck of 32 cards, consisting of the ranks 7, 8, 9, 10, J , Q , K , A in each of the four suits (spades, hearts, diamonds, clubs).

Define the random variable X as follows:

$$X = \begin{cases} -1 & \text{if the card drawn is a 7, 8, 9 or 10,} \\ 1 & \text{if the card drawn is a Jack, Queen, or King,} \\ 2 & \text{if the card drawn is an Ace.} \end{cases}$$

1. Find the probability density function of X (0.5pt).
2. Compute the expected value $E[X]$ and the variance $\text{Var}(X)$ (0.5pt).
3. Compute $P(X \geq 0)$ (0.5pt).
4. Let $Y = 2X - 1$. Compute the expected value $E[Y]$ (0.5pt).

Some help

$$\int x^n dx = \frac{x^{n+1}}{n+1} \quad (a \neq 0)$$

$$\int e^{ax+b} dx = \frac{1}{a} e^{ax+b} \quad (a \neq 0)$$

$$\int fg' dx = fg - \int f'g dx$$

where f and g are functions of x

$$\frac{d}{dx}(\log x) = \frac{1}{x} \quad x > 0$$

$$\frac{d}{dx}\left(\frac{1}{x^n}\right) = -\frac{n}{x^{n+1}} \quad (x \neq 0)$$

$$\log(ab) = \log a + \log b$$

$$\log(a^n) = n \log a$$

$$\text{Corr}_{X,Y} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

Standard Normal (Z) Table:

The table below gives $P(Z < z)$ for a standard normal variable.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.535
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.575
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.614
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.651
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.687
0.5	0.6915	0.6944	0.6973	0.7003	0.7032	0.7061	0.7090	0.7119	0.7148	0.717
0.6	0.7257	0.7284	0.7314	0.7340	0.7367	0.7394	0.7420	0.7446	0.7472	0.749
0.7	0.7580	0.7606	0.7632	0.7658	0.7684	0.7709	0.7734	0.7759	0.7784	0.780
0.8	0.7881	0.7907	0.7932	0.7957	0.7981	0.8005	0.8029	0.8053	0.8076	0.810
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.838
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.862
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.901
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.917
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.931
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.944
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.954
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.963
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.970
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.976
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.981
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.985
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.989
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.991
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.993
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.995
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.996
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.997
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.998
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.998